Classifying NFL Coach Career Movement: A Machine Learning Approach to Analyzing Racial Disparities in Coaching Hierarchies

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EECE 5644: Introduction to Machine Learning
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Abstract

The National Football League (NFL) has long faced criticism for the underrepresentation of minority coaches in leadership positions. Despite corrective policies such as the Rooney Rule being in place for nearly two decades, persistent hiring disparities remain. This paper applies multi-class classification techniques to predict the season-to-season career trajectory of NFL coaches—specifically whether they are promoted, demoted, laterally moved, or exit the league. We train and evaluate five machine learning models: K-Nearest Neighbors (KNN), Random Forest, Decision Tree, Naive Bayes, and Linear Discriminant Analysis (LDA). Our dataset combines NFL team performance statistics from 2003–2023 (672 samples across 35 features) with a comprehensive coach list spanning 1989–2025 that includes racial demographic information (14,745 samples across 7 categorical features). We address significant class imbalance through oversampling and cost-sensitive learning, and perform hyperparameter tuning via grid search on our best-performing tree-based models. Our results reveal persistent challenges in classifying minority movement classes, particularly promotions, and highlight the role of engineered features such as statistical positionality and league churn rate in model performance. These findings contribute to a data-driven understanding of where barriers to advancement may exist within the NFL coaching pipeline.

1. Introduction

The question of racial equity in professional sports leadership has attracted sustained attention from researchers, policymakers, and the public alike. In the NFL, fewer than 25%of players are white, yet at the start of the 2024 season, more than 80% of head coaches were white (Farnell et al. 2025). This stark disparity persists nearly two decades after the introduction of the Rooney Rule in 2003, an affirmative action policy requiring teams to interview minority candidates for head coaching vacancies.

This project seeks to classify the career trajectory of NFL coaches on a season-to-season basis using team statistical features, and then observe whether meaningful trends emerge along racial lines. The classification task assigns each coach-season observation one of four movement labels: *Promotion*, *Sideways* (lateral move or same role), *Demotion*, or *Exit* (departure from the dataset). By framing coaching career dynamics as a supervised learning problem, we aim to identify which features most strongly predict movement and whether the models reveal structural patterns that correlate with race.

Several factors make this task challenging. First, team-level statistics must be individualized to approximate a coach's specific contribution which is difficult given that coaching performance is inherently collaborative. Second, the dataset contains much more detailed offensive statistics than defensive. Third, the dataset exhibits significant class imbalance since the vast majority of coaches remain in their current role from season to season. Lastly, the promotion class (while not being the smallest of the minority classes) is consistently misclassified at much higher rates than other classes across all algorithms.

2. Background

2.1 Machine Learning Principles

This work applies supervised multi-class classification, in which a model learns a mapping from input features to one of the four discrete output categories. We employ five distinct algorithms:

K-Nearest Neighbors (KNN) is a non-parametric, instance-based method that classifies a sample by majority vote among its k nearest training examples in feature space. It is a lazy learner in the sense that most computation is deferred to the prediction. **Decision Trees** recursively partition the feature space along simple feature-based decision boundaries that optimize a specified criterion like information gain or Gini impurity. **Random Forest** is a meta estimator that fits multiple decision tree classifiers on various dataset sub-samples and uses averaging to improve predictive accuracy while controlling overfitting. This is one of two models we deploy that were not explicitly covered dur-

ing class. **Naive Bayes** applies Bayes' theorem under the assumption that features are conditionally independent given their class label. **Linear Discriminant Analysis** (LDA) identifies a linear boundary to separate classes by assuming each group follows a Gaussian distribution with a shared covariance. It assigns new data points to the class with the closest mean as measured by Mahalanobis Distance, effectively maximizing class separation. This is the second model we deploy that was not explicitly covered during class.

2.2 Addressing Class Imbalance

Class imbalance is a pervasive issue in real-world classification tasks and is particularly acute in our setting. We observe that the majority class (Sideways) dwarfs the minorities, comprising nearly 72% of all features. The promotions and demotions make up just over 5% and 3% respectively while exits are almost 20%—greater than promotions and demotions combined but still a fraction of the lateral moves. We employ two strategies to mitigate this:

Oversampling generates synthetic minority-class examples (via SMOTE) to equalize class representation in the training set. This generally tends to improve recall for minority classes at the cost of precision and overall accuracy.

Cost-sensitive learning modifies the loss function to impose higher penalties for misclassifying minority-class samples. In tree-based methods (i.e. Decision Tree and Random Forest), this is achieved by setting `class_weight='balanced'`, which inversely weights classes by their frequency.

3. Related Work

Farnell et al. (2025) conducted a comprehensive study of the NFL coaching pipeline spanning 1989–2020. Using logit and multinomial logit models, they tracked coach movements across three hierarchical levels: head coach, coordinator, and position coach. Their principal findings were as follows: (1) black head coaches remain substantially underrepresented relative to the player population; (2) race is not a statistically significant factor in explaining promotions to head coach or movements of coordinators; (3) conditional on observable performance factors, black positional offensive coaches are significantly less likely to be promoted, with a relative risk ratio of 0.216—though this disadvantage disappears once a coach reaches the quarterback coaching position; and (4) the underrepresentation of minority head coaches appears attributable to barriers earlier in the coaching pipeline rather than the perceived discrimination at the head coaching hiring stage.

Their conclusion that the pipeline problem begins at the position-coach level rather than the coordinator or head-coach level ran counter to my preconceived notion of the failures of the Rooney Rule. As a result, it directly motivated my project. In this project, I aim to test whether machine learning classifiers, trained on team performance statistics and engineered coach-level features, can produce meaningful predictions—allowing us to examine differences across racial boundaries.

Our approach deviates from Farnell et al. in several key ways. First, we redefine the movement categories: we treat lateral moves as equivalent to remaining in the same position and classify all departures from the following season's dataset as exits, rather than distinguishing between sideways and downward moves. Second, we operate on a year-to-year classification basis using team statistical features rather than econometric modeling with logistic regressions. Third, we apply five distinct machine learning algorithms with systematic hyperparameter tuning on our best performing models , whereas Farnell et al. relied on parametric statistical models.

4. Problem Formulation and Dataset

4.1 Task Definition

We formulate this as a **multi-class classification** problem. Given a set of features describing a coach and their team's performance in season t, the goal is to predict the coach's *Movement* in the transition from season t to season $t+1$. Movement is an ordinal variable with four classes:

- **Promotion (3):** Coach moves to a higher tier in the hierarchy (e.g., position coach to coordinator).
- **Sideways (2):** Coach remains in the same role or makes a lateral move (treated as equivalent).
- **Demotion (1):** Coach moves to a lower tier (e.g., head coach to coordinator).
- **Exit (0):** Coach departs from the dataset entirely.

4.2 Data Sources

We draw on two datasets:

NFL Team Data (2003–2023). Sourced from Kaggle and the `nflreadpy` library, this dataset contains 672 samples (21 seasons $\times$ 32 teams) with 35 features, of which 2 are categorical (team and year) and the remainder are numerical team performance statistics.

Coach List with Race. Sourced from Dr. Alexander Farnell directly at my personal request, the dataset used for the paper contains 14,745 samples covering every NFL coach at the top three tiers (head coach, coordinator, position coach) from 1989–2025, with 7 categorical features including the coach's name, role, team, and race.

We conducted a left-wise merge of the Coach List onto the NFL Team Data dataset. This merge was done on the features team and year from the respective datasets such that a team's statistics for a given season were applied to every

coach on their roster. The Coach List dataset used abbreviations (some of which were non-standard) which required us to use both the nflreadpy library's `load_teams()` function as well as our own `get_historical_name` helper function that accounted for differences in team abbreviations between the library and our Coach List (e.g., NEW for the New England Patriots instead of NE).

4.3 Features

Inherent features include categorical variables (coach name, year, role, team) and numerical team statistics drawn from `nflreadpy` (e.g., points scored, yards gained, turnovers).

Engineered features include:

- **Tier:** Ordinal encoding of a coach's hierarchical position. For our original test, we opted to use the standard coaching hierarchy (Head Coach = 3, Coordinator = 2, Position Coach = 1).
- **Movement:** The target variable, computed by comparing a coach's current tier to their tier in the following season (Promotion = 3, Sideways = 2, Demotion = 1, Exit = 0).
- **CoachID:** A unique identifier concatenating coach name with date of birth to disambiguate coaches with similar names (e.g., the Harbaugh brothers, the LaFleur brothers). The edge case that potentially stood as a threat was the Ryan twins (Rex and Rob) who would have the same Name attribute of R. Ryan. It appeared that Rob (the less notable of the two) never appeared in the Coach List dataset for some reason which meant we did not need to distinguish between the two.
- **Experience:** The cumulative number of seasons coached, computed by counting prior appearances of the CoachID across all years in the dataset (including years prior to 2003, before rows were dropped for lack of team statistics).
- **Statistical Positionality:** A measure attempting to capture how well a coach performed in their specific role relative to their peers in a given season. It uses our `relevant_features` map to determine what features to check for a specific coaching position. It groups coaches by position and year then calculates the coach's percentile ranking (inverted for metrics where lower is better like `points_opp`) for the given feature—averaging across the relevant features for the position.
- **Previous League Churn Rate:** A feature capturing the historical rate of coaching turnover in the league. To avoid data leakage, we only look at the churn rate for the previous season (e.g., for the 2020 season we observe churn from 2018 to 2019). This feature is applied to every coach in a given year.

Figure 1: NFL Coaching Churn Rate

4.4 Target Variable

The target variable is **Movement**, which encodes the transition outcome for each coach between consecutive seasons. The class distribution is heavily imbalanced, with the Sideways class dominating and Demotion constituting the smallest class.

5. Methodology

5.1 Exploratory Data Analysis

We found no meaningful results in the summary statistics for the merged dataset. Thus, we began with alternative visualizations to understand the structure and distribution of our data (with an intended goal of interrogating the imbalance). A bar plot of coach roles against movement classes (e.g., Figure 2) revealed misalignment in the originally proposed coaching tiers. For example, special teams coaches were found to almost never receive promotions, while offensive and defensive coordinators were more likely to be demoted than head coaches: consistent with the higher volatility of coordinator positions documented in the literature. A second set of bar plots examined which coaches appeared most frequently within each movement class (e.g., Figure 3) , revealing that promotions are heavily dominated by quarterback coaches. This finding aligns with Farnell et al.'s (2025)

Figure 2: Special Teams Coordinator Movement Outcomes

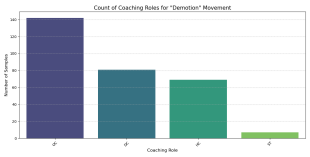

Figure 3: Demotions by Coach Role

observation that the quarterback coach role serves as a critical stepping stone in the coaching hierarchy, with 23% of all head coaches in their sample having held this position at some point.

5.2 Feature Selection and Correlation

To refine our feature set, we analyzed the correlation between each numerical variable and the target movement class (**Figure 3**). Our selection process prioritized features with a demonstrable statistical relationship to coaching transitions while adhering to three critical constraints. First, we eliminated redundant features to reduce multicollinearity. Second, we excluded variables that were overly position-specific (such as `pass_net_yds_per_att`)which would have limited the model's ability to generalize across the entire coaching staff. Finally, we removed any features that risked data leakage by containing information from future seasons (such as `Next_Tier`) ensuring the model's predictions remained grounded in contemporary data.

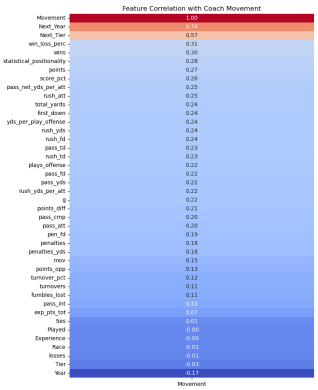

Figure 4: Feature Correlation with Target

5.3 Data Preprocessing

Racial Filtering, All coaches whose race was neither "B" nor "W" in the dataset were filtered out. Since our main problem was already a multi-class classification, we thought it would be too distracting to deal with the noise of coaches from other races in our evaluation. Because this was a small minority of the data samples, we were not terribly concerned with the impact it would have.

Categorical encoding. Race was encoded as a binary variable. The Tier feature was ordinally encoded (Position Coach = 1, Coordinator = 2, Head Coach = 3).

Temporal filtering. All observations prior to 2003 were dropped because team performance statistics were unavailable for those seasons. Coach experience was computed before this filtering step to preserve the full career history.

Dataset merging. The two datasets were merged by mapping team statistics to each coach based on team and year. Since the coach list dataset did not include full team names, we mapped team names to abbreviations using the internal `nflreadpy` mapping. Additional manual corrections were required for teams whose abbreviations did not match across datasets—either because of differing conventions or franchise relocations (e.g., STL → LAR for the St. Louis/Los Angeles Rams).

Feature scaling. Numerical features were standardized prior to training models.

5.4 Model Selection

We trained five classifiers:

1. K-Nearest Neighbors (KNN)
2. Naive Bayes
3. Decision Tree
4. Random Forest
5. Linear Discriminant Analysis (LDA)

All models were implemented using `scikit-learn`. We established a baseline by training each model with default hyperparameters on the imbalanced dataset. The baseline models achieved reasonable overall accuracy but exhibited poor performance on the minority classes, particularly Promotion.

6. Results and Evaluation

6.1 Baseline Performance

All five models were first evaluated on the imbalanced dataset using default hyperparameters. The models achieved reasonable overall accuracy, driven primarily by correct classification of the majority Sideways class. However, F1 scores for the minority classes—particularly Promotion—were substantially lower across all models, confirming the expected difficulty of identifying rare career transitions from aggregate team statistics.

Since Random Forest was our best performing classifier in terms of accuracy and macro F1 score, we observe its confusion matrix. We can see that the problematic Promotion class is never classified accurately whereas Sideways boasts 1296 true positives. The revelations from the baseline performance prompted us to employ techniques that can address the imbalance

6.2 Effect of Oversampling

Applying oversampling to equalize the training class distribution produced an expected trade-off: overall accuracy dropped while the macro F1 score improved.

Linear Discriminant Analysis, one of the better performing baseline models, was impacted severly by the over-

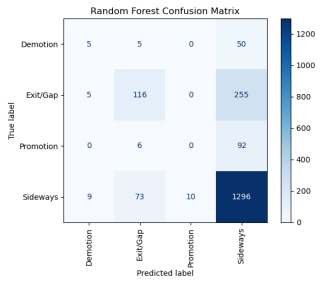

Figure 5: Random Forest Baseline Confusion Matrix

sampling which resulted in an approximate 43% drop in overall accuracy.

Even for a more robust class like **K-Nearest Neighbors** the accuracy drop was considerable. That being said, we can see in figures 6 and 7 that the average 9% increase in F1 score for each of the minority classes was a benefit of oversampling on the model.

```
==================== KNN (SMOTE) ====================
              precision    recall  f1-score   support

    Demotion       0.18      0.57      0.28        60
    Exit/Gap       0.31      0.46      0.37       376
   Promotion       0.07      0.22      0.11        98
    Sideways       0.81      0.51      0.62      1388

    accuracy                           0.48      1922
   macro avg       0.34      0.44      0.34      1922
weighted avg       0.65      0.48      0.54      1922
```

Figure 6: KNN (SMOTE) Classification Report

```
======================== KNN ========================
              precision    recall  f1-score   support

    Demotion       0.21      0.13      0.16        60
    Exit/Gap       0.42      0.28      0.34       376
   Promotion       0.00      0.00      0.00        98
    Sideways       0.77      0.88      0.82      1388

    accuracy                           0.70      1922
   macro avg       0.35      0.33      0.33      1922
weighted avg       0.64      0.70      0.66      1922
```

Figure 7: KNN Baseline Classification Report

6.3 Effect of Cost-Sensitive Learning

We also explored cost-sensitive learning on the tree-based models by using balanced class weights. The impact of this was a bit opaque as the **Decision Tree** model had improved recall for Promotions while the other three classes all saw decreases. For this model the macro F1 dropped from 0.36 to 0.34 whereas Random Forest saw an increase from 0.34 to 0.35.

6.4 Hyperparameter Tuning Impact

Since the tree-based models showed the most consistency in terms of performance (as measured by macro F1 score and accuracy) across the differences in class imbalance, we chose them as our two candidates for hyperparameter tuning. The goal was to determine which we would choose as our favored model to complete our final evaluation on with additional feature engineering. We deployed grid-search with five-fold cross validation.

Random Forest. Grid search identified the following optimal parameters: `class_weight=balanced_subsample`, `max_depth=25`, `min_samples_leaf=5`, `min_samples_split=2`, `n_estimators=50`. The best cross-validated macro F1 score was 0.4235. Relative to the default hyperparameter configuration, the tuned model adopted a balanced subsample class weighting (from `None`), reduced maximum depth to 25 (from unlimited), and increased the minimum sample leaves to 5 (from 2).

Decision Tree. The tuned model retained the Gini criterion (default) but adopted a maximum depth of 15 (from unlimited), which helped reduce overfitting while maintaining classification performance. The best cross-validated F1 score was 0.3592

Random Forest clearly showed more substantial improvement as a result of the hyperparameter tuning whereas **Decision Tree's** performance remained within a margin of error of its original macro F1 scores. We will proceed with Random Forest for a final evaluation with its optimal hyperparameters.

7. Final Evaluation

7.1 Additional Feature Engineering

After landing on Random Forest (with its designated hyperparameters) as the model for our final evaluation, I opted to make some additional changes to the dataset to address the pervasive class imbalance. The first change I made was to drop all Special Teams coaches from the dataset since they have a negligible ($\leq 1\%$) likelihood of promotion or demotion based on the data. This was actually a strategy employed by Farnell et. al in their research but I thought I would first try experimenting with them included since I knew of anecdotal instances of Head Coaches who had a background in Special Teams (e.g., John Harbaugh, Joe Judge). These instances proved to be quite anecdotal and since every team, in every season, in every year of our dataset had a special teams coordinator, it was introducing a great amount of noise into our data. The result was a reduction of total samples by over 37%.

I also decided to extend the coach tier hierarchy from 3 to 5 tiers to better encapsulate the hierarchy that exists at the position coach level. The majority class went from 72% of the distribution to about 69% whereas the smallest two minority classes each saw a 1% increase. While the difference may appear small, we hope that this coupled with the improved model will give us results that we can begin to draw

conclusions from.

7.2 Evaluation

The final tuned Random Forest model demonstrated a significant improvement in identifying minority class transitions, which were largely obscured in baseline iterations. By utilizing the 5-tier hierarchy and addressing class imbalance, the model achieved a more nuanced predictive profile across all career movements.

7.2.1 Model Performance

While the overall accuracy settled at 64%, the model's utility is better reflected in its **Macro-F1 score of 0.43**. Key performance metrics by class include:

- **Demotion:** Achieved a high **recall of 0.53**, indicating strong sensitivity to downward mobility features.
- **Promotion:** Improved to a **precision of 0.20**, marking a successful departure from baseline models that failed to predict any promotions.
- **Sideways:** Maintained stability with an **F1-score of 0.76**, reliably identifying the majority class without excessive false positives.

Final Optimized Model Performance (Test Set):

	precision	recall	f1-score	support
Demotion	0.23	0.53	0.33	73
Exit/Gap	0.47	0.43	0.45	351
Promotion	0.20	0.18	0.19	114
Sideways	0.78	0.74	0.76	1232
accuracy			0.64	1770
macro avg	0.42	0.47	0.43	1770
weighted avg	0.66	0.64	0.64	1770

Figure 8: Final Evaluation Model Performance

8. Analysis and Interpretation

8.1 Feature Importance

Analysis of feature importance in the Random Forest model revealed that several engineered features ranked among the most predictive. *Statistical Positionality* and *Previous League Churn Rate* ranked highly by both mean decrease in impurity and permutation importance, suggesting that a coach's positional context and the broader coaching labor market dynamics are informative signals for predicting career movement.

Other engineered features, such as *Experience*, proved to be of marginal utility. This likely reflects the coarseness of the feature: a simple count of seasons coached may not capture the nuanced aspects of coaching tenure that influence career outcomes, such as the prestige of prior roles, experience is current role, or the performance trajectory over a career.

--- Feature Importance Comparison ---

	Feature	MDI_Importance	Permutation_Importance	Perm_Std
0	Tier	0.229335	0.068924	0.010740
1	prev_league_churn	0.110586	0.030387	0.006201
2	score_pct	0.108925	0.010076	0.004087
3	statistical_positionality	0.151082	0.009885	0.009028
4	Race	0.023465	0.006936	0.005270
5	Experience	0.104881	0.003941	0.005187
6	wins	0.074658	0.001709	0.006657
7	mov	0.054643	0.001087	0.002679
8	points_diff	0.142424	0.000559	0.008452

Figure 9: Feature Importance Comparison

8.2 Disparities Across Racial Boundaries

The final Random Forest model reveals a notable gap in predictive performance across racial lines. While the model achieved an overall accuracy of 64%, it correctly classified 71.25% of Black coaches compared to only 59.53% of White coaches. This suggests that the structural features used (particularly *Tier* and *Statistical Positionality*) provide a more consistent signal for the career paths of Black coaches, whereas the movements of White coaches may be influenced by a higher degree of "noise" or unobserved factors like informal networking.

Interestingly, *Race* yielded a low permutation importance (0.0069) compared to *Tier* (0.0689), indicating the model is likely identifying the "proxy" effects of race rather than race as a direct causal driver. To further investigate these influences, future research should employ fairness-aware regularization to prevent the replication of historical biases and conduct a longitudinal "time-in-grade" analysis. Such steps would help determine if minority coaches are held to higher statistical thresholds or longer tenures before reaching the same promotional milestones as their counterparts.

8.3 Critical Reflection

The fundamental challenge of this project can be summarized as follows: the features available to us are team-level performance statistics, but the prediction target—an individual coach's career movement—is influenced by factors that team statistics cannot fully capture. Coaching hires and promotions depend on network effects, interview performance, front-office relationships, and subjective evaluations that unfortunately cannot be encapsulated using these statistics.

The persistent difficulty in classifying promotions suggests that the signal distinguishing promoted coaches from those who remain or exit is either very weak in the available features or confounded by unobservable variables. Future work could benefit from incorporating features that more directly capture individual coaching quality, such as unit-level performance metrics, more sophisticated experience features (e.g., experience in a specific role), or player performance trajectories under specific coaches.

9. Real-World and Social Impact

9.1 Applications

This work has potential applications for NFL front offices seeking to develop data-informed approaches to coaching evaluation and hiring. A predictive model of coaching career trajectories could supplement traditional scouting and interview processes by identifying coaches whose statistical profiles suggest readiness for promotion. More broadly, the methodology could be adapted to study career advancement patterns in other hierarchical organizations where demographic disparities in leadership persist.

9.2 Ethics and Bias

The data reveals a clear structural bias against minority coaches in career advancement, consistent with the findings of Farnell et al. (2025). Our classification models inherit and potentially amplify these biases: if historical data reflects discriminatory hiring patterns, a model trained on this data may learn to reproduce those patterns rather than identify meritorious candidates.

Determining where and how to correct for such bias is the difficult part. Simply removing race as a feature does not eliminate the problem, as other features (e.g., positional role, which correlates with race due to historical patterns) may serve as proxies. Any deployment of such models in real hiring contexts would require careful fairness auditing, transparent documentation of model limitations, and human oversight to prevent algorithmic perpetuation of existing inequities.

10. Conclusion

This project applied multi-class classification to predict the season-to-season career trajectories of NFL coaches using team performance statistics and engineered features. We trained and evaluated five models—KNN, Naive Bayes, Decision Tree, Random Forest, and LDA—and found that tree-based methods consistently performed best, particularly after hyperparameter tuning and class-imbalance mitigation.

Our analysis confirms that team-level statistics are insufficient for predicting career-level movement, with the Promotion class proving especially challenging to classify. Engineered features such as Statistical Positionality and Previous League Churn Rate contributed meaningfully to model performance, while simpler features like raw experience counts offered limited predictive value.

The broader implications of this work align with prior research: the barriers to minority coaching advancement appear to operate at levels of the hierarchy that are difficult to address through top-level policies alone. Machine learning approaches can help quantify and characterize these structural patterns, but meaningful change will require interventions that address the root causes of disparate career trajectories in the coaching pipeline.

Future work should explore richer feature sets that better capture individual coaching performance, investigate temporal dynamics in coaching markets (e.g., the "Sean McVay effect" favoring young offensive-minded coaches), and apply fairness-aware machine learning techniques to develop models that can identify talent equitably across demographic groups.

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